MATH317 · NUMERICAL METHODS

Homework 2 Solutions

Problems 2.1 · Floating-point

Muharib

01 4(f): representing 64.37109375

Determine the single-precision and double-precision machine representation of 64.37109375.

We start with this one because it is easier: its fraction terminates.

Sign: positive → s = 0

$64 = (100)_8 = (1000000)_2$

$.37109375 \times 8 = 2.96875 \rightarrow 2$

$.96875 \times 8 = 7.75 \rightarrow 7$

$.75 \times 8 = 6.0 \rightarrow 6$

$.37109375 = (.276)_8 = (.010111110)_2$

$64.37109375 = (1000000.01011111)_2 = 1.00000001011111 \times 2^6$

Single precision

$c = 6 + 127 = 133 = (10000101)_2$

0 | 10000101 | 00000001011111000000000

0100 0010 1000 0000 1011 1110 0000 0000

$= [4280BE00]_{16}$

Double precision

$c = 6 + 1023 = 1029 = (10000000101)_2$

0 | 10000000101 | 00000001011111 000...0

0100 0000 0101 0000 0001 0111 1100 0000 ... 0000

$= [405017C000000000]_{16}$

02 4(c): representing −9876.54321

Determine the single-precision and double-precision machine representation of −9876.54321.

1 Sign and integer part

```
Negative   →   s = 1
```

```
9876 ÷ 8 = 1234   r 4
```

```
1234 ÷ 8 = 154    r 2
```

```
 154 ÷ 8 = 19     r 2
```

```
  19 ÷ 8 = 2      r 3
```

```
   2 ÷ 8 = 0      r 2
```

$$9876 = (23224)_8 = (10\ 011\ 010\ 010\ 100)_2 = (10011010010100)_2$$

2 Fraction part (it does not terminate)

```
.54321 × 8 = 4.34568  →  4
```

```
.34568 × 8 = 2.76544  →  2
```

```
.76544 × 8 = 6.12352  →  6
```

```
.12352 × 8 = 0.98816  →  0
```

```
.98816 × 8 = 7.90528  →  7
```

```
.90528 × 8 = 7.24224  →  7      ...
```

$$.54321 = (.426077...)_8 = (.100\ 010\ 110\ 000\ 111\ 111\ ...)_2$$

The fraction never ends, so we write enough bits and round at the end of the f field.

3 Normalize

$10011010010100.100010110000111111...$

$= 1.00110100101001000101100001111111... \times 2^{13}$

Single precision

$c = 13 + 127 = 140 = (10001100)_2$

f = first 23 bits = 00110100101001000101100 (next bit is 0, so no round-up)

1 | 10001100 | 00110100101001000101100

1100 0110 0001 1010 0101 0010 0010 1100

$= [C61A522C]_{16}$

Double precision

$c = 13 + 1023 = 1036 = (10000001100)_2$

f = first 52 bits of the fraction

1 | 10000001100 | 0011010010100100010110000111111...

$= [C0C34A4587E7C06E]_{16}$

Both hex values were checked by computer. In double precision the last digits depend on far-away fraction bits, so show the method and at least the leading digits.

03 8(c): hex to decimal

Determine the decimal number that has the machine representation $[494F96A0]_{16}$.

```
 4    9    4    F    9    6    A    0
```

```
0100 1001 0100 1111 1001 0110 1010 0000
```

```
s = 0        c = 10010010 = 146        f = 10011111001011010100000
```

```
m = 146 − 127 = 19
```

$$\text{value} = (1.10011111100101101 01)_2 \times 2^{19}$$

$$= (11001111110010110101 0)_2$$

$$= 850282$$

04 11: a small floating-point system

Enumerate the set of numbers in the floating-point number system that have binary representations of the form $\pm(0.b_1b_2)_2 \times 2^k$, where $k \in \{-1, 0\}$.

The mantissa has four possible values and the exponent has two:

$(0.b_1b_2)_2$	Value	$\times 2^{-1}$	$\times 2^{0}$
0.00	0	0	0
0.01	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{4}$
0.10	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{2}$
0.11	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$

$$\{\ 0,\ \pm\frac{1}{8},\ \pm\frac{1}{4},\ \pm\frac{3}{8},\ \pm\frac{1}{2},\ \pm\frac{3}{4}\ \}$$

Eleven numbers in total. The value $\frac{1}{4}$ appears twice but is counted once.

05 15: the error in underflow

In the case of machine underflow, what is the relative error involved in replacing a number x by zero?

$$\text{Relative error} = \frac{|\ x - 0\ |}{|\ x\ |} = 1$$

$$= 100\ \%$$

All significant digits are lost, however small x is.

06 18: when is fl(1 + x) = 1?

In a five-decimal machine that correctly rounds numbers to the nearest machine number, what real numbers x will have the property fl(1.0 + x) = 1.0?

Look at the machine numbers next to 1 on this five-digit machine:

$$1 = 0.10000 \times 10^{1}$$

$$\text{Next number above:} \quad 0.10001 \times 10^{1} = 1.0001 \qquad (\text{gap} = 10^{-4})$$

$$\text{Next number below:} \quad 0.99999 \times 10^{0} = 0.99999 \qquad (\text{gap} = 10^{-5})$$

`1 + x` rounds to 1 when it is closer to 1 than to either neighbour, that is, within half a gap on each side:

$$0.999995 \;<\; 1 + x \;<\; 1.00005$$

$$-0.5 \times 10^{-5} \;<\; x \;<\; 0.5 \times 10^{-4}$$

Exactly at the endpoints the result depends on the tie-breaking rule.

07 21: rounding $2^{-1} + 2^{-26}$

(Continuation) What is the relative roundoff error when we round off $2^{-1} + 2^{-26}$ to get the closest machine number?

Assume single precision (a 24-bit mantissa including the leading 1).

$$x = 2^{-1} + 2^{-26} = 2^{-1}\,(1 + 2^{-25})$$

Machine numbers near 2^{-1} are spaced $2^{-1} \times 2^{-23} = 2^{-24}$ apart

2^{-26} is less than half of that spacing (2^{-25}) → $fl(x) = 2^{-1}$

$$\text{Relative error} = \frac{|\,x - fl(x)\,|}{|\,x\,|} = \frac{2^{-26}}{2^{-1} + 2^{-26}}$$

$$\approx 2^{-25} \approx 2.98 \times 10^{-8}$$

The problem is marked "Continuation", so it follows an earlier problem in the book. Single precision is assumed here; check the wording of that earlier problem.

08 38: subtracting two close numbers

Consider a computer that uses five-decimal-digit numbers. Show that if x = 0.5321487513 and y = 0.5321304421, then the operation fl(x) − fl(y) involves a large relative error. Compute it.

$fl(x) = 0.53215$ $\qquad$ $fl(y) = 0.53213$

$fl(x) - fl(y) = 0.00002$

Exact: $x - y = 0.0000183092$

$$\text{Relative error} = \frac{|\,0.0000183092 - 0.00002\,|}{0.0000183092}$$

$$= \frac{0.0000016908}{0.0000183092}$$

$$\approx 0.092 \quad \text{(about 9.2 \%)}$$

Why is the error large?

Each number alone was rounded with a tiny error. After subtracting, the matching digits cancel and the error remains, now 9% of the result. This is **Loss of significance**.